Ubuntu Naturals Lagos: cobot inspection cell, IIoT architecture (worked example)

Worked example, DigiSkills | 30 September 2026 | Stage 4 encapsulation plant, Lagos, Nigeria

For the plant manager
The twin predicts 11 protective stops and 469 trays in a 4-hour shift; the log shows 22 and 448. This design gives the twin the three signals behind the gap.

The architecture

→ Data flow (C) Protocol, listed right (1) One lidar stop, data to decision ⬚ Hard-wired safety

Sensor / device (IIoT)	Safety lidar (speed and separation monitoring): S_p 1700 mm at 400 mm/s Vision camera, vacuum switch, safety PLC, production PLC (B) PROFINET: production PLC to cobot and camera (A) **Hard-wired stop: lidar, safety PLC, cobot drive**	**1 Data** Safety PLC state STOPPED; no AGV in the lane; lidar window flag set
	(C) **OPC UA, read-only** (D) **AGV fleet manager REST, polled**	
Edge / gateway (IIoT)	Gateway on the cell UPS: joins each stop to the AGV position and lidar diagnostics Store-and-forward, sequence-numbered; HMI stays local	**2 Information** A nuisance trip: nobody and no AGV inside S_p
	(E) **MQTT over TLS**	
Network (Cybersecurity)	OT VLAN and firewall; TLS from the gateway to the broker The broker keeps the session through link drops The safety stop is not on this network	**3 In transit** Sent with its sequence number, first after an outage
	(E) **Sparkplug B, QoS 1**	
Cloud / platform (Cloud, Simulation)	Group platform event store: stops, trays, AGV arrivals The cell twin reruns each shift on the logged AGV times, trips and restarts	**4 Knowledge** The rerun shows 8 nuisance trips mid-shift, a term the twin lacks
	(F) **REST API**	
Application (Big Data, System integration)	MES (ISA-95 L3): shift divergence report in 15-min bins NAFDAC stop record; lidar window cleaning order AGV schedule review	**5 Wisdom** Clean the lidar window on a schedule; add trips as a twin input

A protocol at each boundary

- (A) **Lidar, safety PLC, cobot drive: hard-wired, dual channel**
  The stop works with every network down; the IIoT only reads it.
- (B) **Production PLC to cobot and camera: PROFINET**
  Motion and accept / reject results need a deterministic cycle.
- (C) **Both PLCs to the gateway: OPC UA, read-only client**
  One information model for two controllers, with no write rights.
- (D) **AGV fleet manager to the gateway: REST, polled**
  Only the fleet manager records when an AGV entered the lane.
- (E) **Gateway to platform: MQTT over TLS, Sparkplug B, QoS 1**
  Small, irregular messages. The broker keeps the session through link drops, and QoS 1 resends whatever it has not acknowledged.
- (F) **Platform to applications: REST API**
  The twin, the shift report and the NAFDAC export read one store.

Events, not streams: a modelled shift of 469 trays and 11 stops is well under a megabyte, and camera images stay at the edge.

When the grid drops

1. The gateway is on the cell's UPS. A power dip is when the cell stops, and that stop is the event the record is for.
2. The hard-wired stop and the local HMI work through an outage.
3. Each event is stored locally with a sequence number and a timestamp, and deleted only when the broker acknowledges it.
4. On reconnection, safety events go first (the NAFDAC record an auditor asks for), then state changes, then tray counts.
5. A shift with a gap in its sequence numbers is marked incomplete and kept out of the twin.

What the connection lets the twin see

What diverges	Model	Shift log	Signal that shows it	Use in the twin
AGV arrivals	One every 25 min: 9 AGV stops	12 arrivals, bunched late in the shift	Arrival times from the AGV fleet manager	Replaces the fixed AGV interval
Stops with nobody in the zone	None	8 nuisance trips, all mid-shift	Stop record joined to AGV position and window flag	A new input the model lacks
Restart time	15 s	34.0 s mean, 22 stops	Safety PLC state times, STOPPED to RUNNING	Replaces the restart parameter

The restart gap is there from the first stop, so the restart time needs re-measuring. The AGV gap opens late in the shift, so it is a question for the logistics schedule.

How the pillars connect

IIoT → Simulation → Cloud → Big Data → System integration

The IIoT layers collect AGV arrival times; the twin reruns on them each shift on the platform, and the shift report goes into MES. On the separated AGV route the model gives 2 stops and 475 trays, against 11 and 469. In the log all 12 AGV arrivals stopped the cell, so on that shift the reroute would have removed 12 AGV stops, not 9.